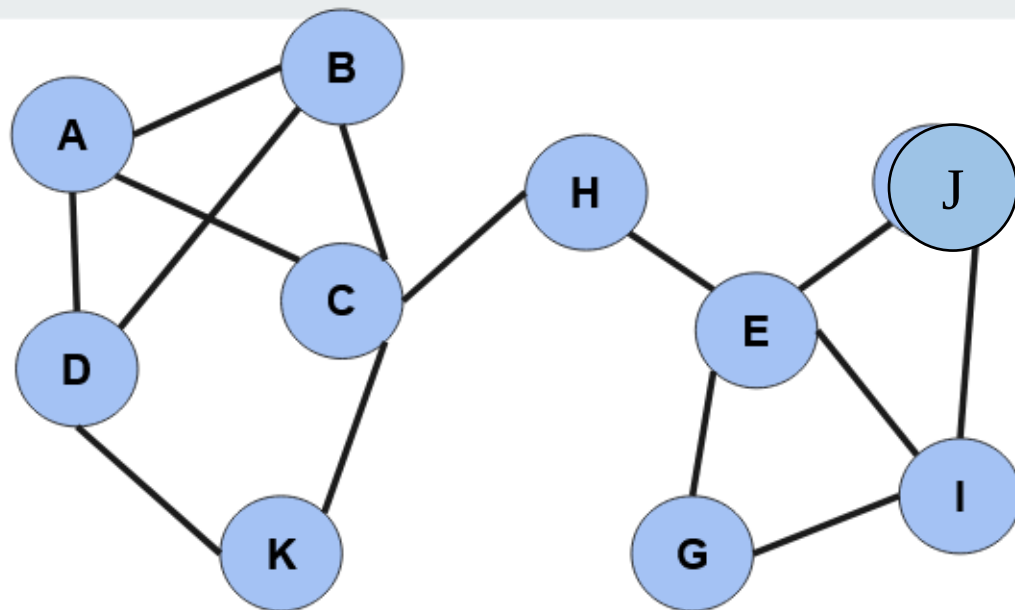


1.



a) What is the order of the graph?

The order of the graph is 10, because there are 10 nodes in the graph.

b) What is the size of the graph?

What is the size of the graph 14, because there are 14 edges in the graph.

c) What is the density of the graph?

$$\frac{m}{\frac{n(n-1)}{2}} = \frac{14}{\frac{10(10-1)}{2}} = \frac{14}{90/2} = \frac{14}{45} = .31$$

$$\frac{2m}{n(n-1)} = \frac{2(14)}{10(10-1)} = \frac{28}{90} = .31$$

OR

Note here that we must multiply the number of edges by 2, or divide the total number of possible connections by 2, so that we can account for the true number of total possible ties. If a node knows every other node in the graph, they will know n-1 other people, because they have to subtract themselves. So one node could know a maximum of 9 other nodes. However, those other nodes can also know them. Because of this mutual knowing, and the fact that the graph is undirected, we need to divide by 2 to account for their mutual knowing in order to get the real number of maximum possible ties, as rather than 90 possible arrows to draw, there is only 45 because of the mutuality represented in the tie.

d) What is the average degree of the graph?

$$\frac{2m}{n} = \frac{2 * 14}{10} = \frac{28}{10} = 2.8$$

Here we take the number of edges and multiply by 2. Note that one edge is contributing 1 degree value to each of the two nodes it is connected to. Thus, to take a shortcut where we don't have to add up the degree of every node, we can take the number of edges and multiply by 2, and then divide by the total number of nodes.

e) Optional Question- Please turn the above graph into a matrix.

f) Is the above matrix an asymmetric, symmetric, or antisymmetric matrix?

Symmetric. The graph is undirected, and this inherent reciprocity will lead to a matrix that is symmetric as the bottom triangle will perfectly mirror the top triangle.

g) What are examples of non-adjacent nodes in the graph?

Any two unconnected nodes, e.g. H and K, or A and I

h) What is the degree set of the above graph?

A	B	C	D	E	G	H	I	J	K
3	3	4	3	4	2	2	3	2	2

i) What is the degree sequence of the above graph?

C	E	A	B	D	I	J	K	H	G
4	4	3	3	3	3	2	2	2	2

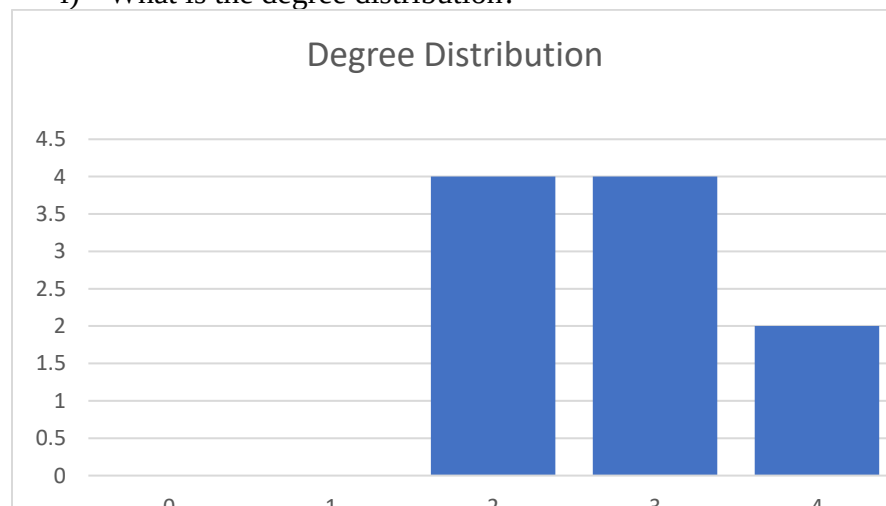
j) What is the degree range?

4-2=2 The degree range is the maximum subtracted from the minimum, providing a measure of the spread within the degree set of the nodes.

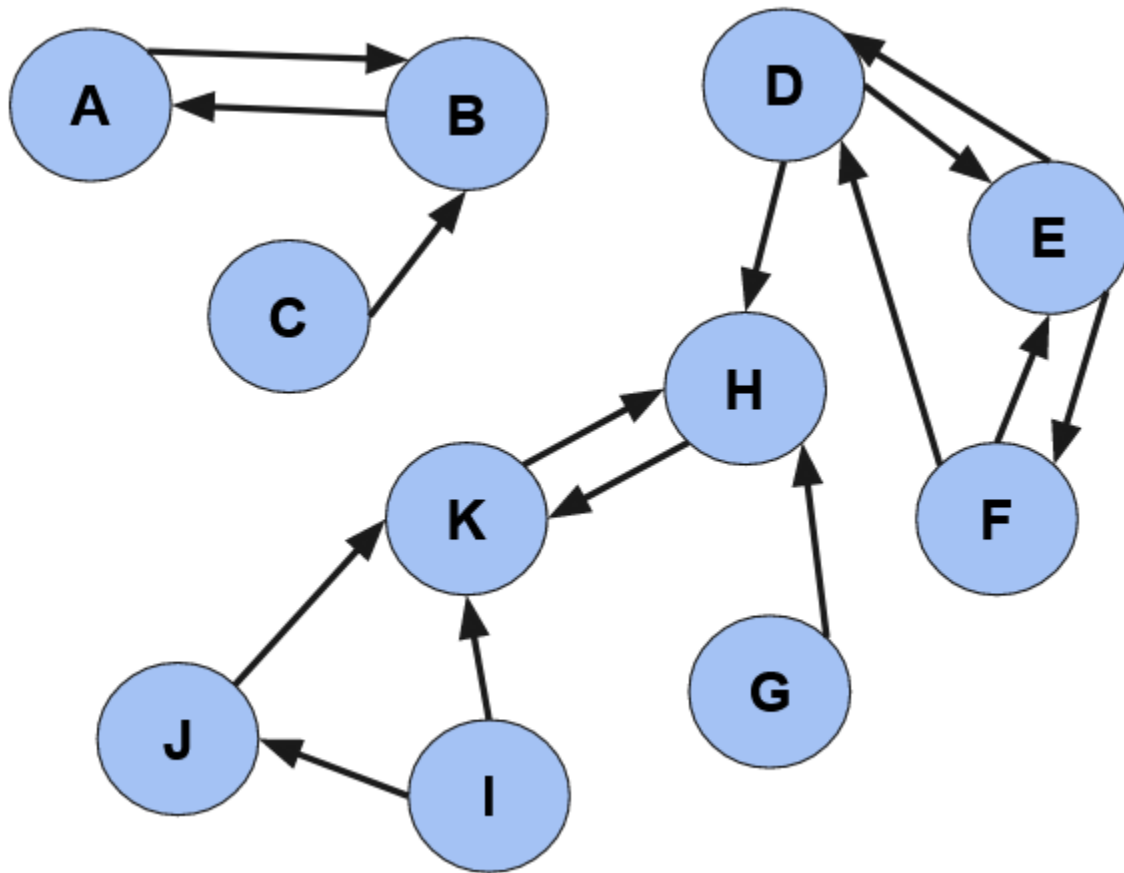
k) What is the minimum and maximum degree?

2 min and 4 max

l) What is the degree distribution?



2.



a) What is the order of the graph?

The order of the graph is 11 because there are 11 nodes.

b) What is the size of the graph?

15 edges, because we simply count how many edges there are, regardless of the directionality.

c) What is the density of the graph?

$$\frac{m}{n(n-1)} = \frac{(15)}{11(11-1)} = \frac{15}{110} = .14$$

The density of the graph is calculated as the number of existing ties divided by the number of possible ties. Because a person can know $n-1$ other people in the network (excluding themselves), then we can have the total number of people times the number of people that they can know. As there is no issue of mutual knowing, as each tie indicates that A knows B and would require another tie for B to know A, then there is no need to correct for the $n(n-1)$ calculation as mutual knowing if in question and would be counted as 2 ties in the above graph.

d) Optional Question- Please turn the above graph into a matrix.

e) Is the above matrix an asymmetric, symmetric, or antisymmetric matrix?

A matrix made from the above graph would be asymmetric, as the ties are directed and may or may not be reciprocated, as evidenced by some of the ties in the graph being reciprocal, and others being unreciprocated.

f) What are examples of non-adjacent nodes in the graph?

Any two unconnected nodes, e.g. A and G, or F and I

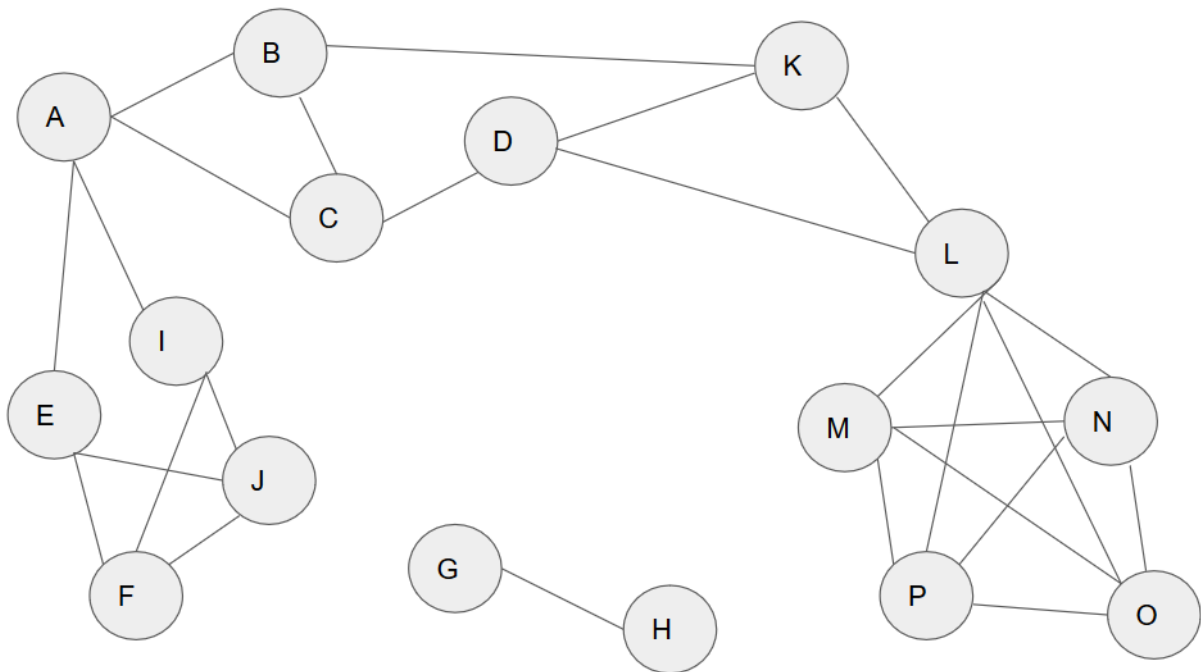
g) What is the average in-degree?

$15/11=1.36$ Each edge in a directed matrix is a tie from one node to another node. Thus, for every edge, there is one in-degree being received. Thus, you can take the number of edges and divide by the number of nodes for the average in-degree.

h) What is the average out-degree?

$15/11=1.36$ Each edge in a directed matrix is a tie from one node to another node. Thus, for every edge, there is one out-degree being sent. Thus, you can take the number of edges and divide by the number of nodes for the average out-degree. It is the exact same answer as G because each edge is both one in-degree and one out-degree.

3)



Concerning connectivity, the above graph is _____. *Disconnected-All nodes can't reach one another.*

Concerning connectivity, the above subgraph $\{V_a, V_b, V_c, V_d, V_e\}$ is _____. *Connected- All nodes can reach one another.*

Concerning connectivity, the above subgraph $\{V_a, V_i, V_k, V_l, V_p\}$ is _____. *Disconnected - All nodes can't reach one another.*

The above graph has _____ components. *2 subgraphs that can't reach other parts of the graph.*

The giant component is _____ connected. *1-connected, would take one node to disconnect the graph*

The diameter of the graph is _____. *6 (Longest possible path in the giant component) FI, IA, AB, BK, KL, LO*

Identify a clique of size 5. *L, M, N, O, P A clique is a maximally connected subgraph*

Identify a clique of size 4. *L, M, N, O A clique is a maximally connected subgraph, so a clique of size 4 can also be part of a clique of size 5.*

Subgraph $\{V_i, V_e, V_f, V_j\}$ is a _____ clique. *2-clique The longest shortest path between two nodes in the subgraph is an n-clique of 2, or a 2-clique.*

Subgraph $\{V_i, V_e, V_f, V_j, V_a\}$ is a _____ clique. *2-clique The longest shortest path between two nodes in the subgraph is an n-clique of 2, or a 2-clique.*

Subgraph $\{V_b, V_c, V_k, V_l\}$ is a _____ clique. *3-clique The longest shortest path between two nodes in the subgraph is an n-clique of 3, or a 3-clique.*

The non-giant component is _____ connected. *It's a dyad, so disconnecting it into more components is not possible.*

Drop nodes G and H when considering the following questions

The above graph is _____ connected. *1-connected, removing one node disconnects the graph.*

The edge cutset of the graph is _____. *DL, KL, or some other set of two edges that would disconnect the graph.*

Removing all 1-cutset nodes from the graph, what are the resulting groups? *EIFJ, MNOP, BCDK. Remove A and L.*

The UV cutset for nodes V_a and V_l is _____. *Some set of edges such as AB and AC which would separate A and L into different components*

Cutsets help measure the theoretic idea of _____. *Groups*